

STAT 542: Statistical Learning

Homework One

Fall 2026, University of Illinois Urbana–Champaign

Due by 11:59 pm on Monday, September 14, 2026

Total Points = 75

1. Consider a clinical test for cancer that can yield either a positive (+) or negative (-) result. Suppose that a patient who truly has cancer has a 1% chance of slipping past the test undetected. On the other hand, suppose that a cancer-free patient has a 5% probability of getting a positive test result. Suppose also that 2% of the population has cancer. Suppose that a patient who has taken the test **twice** has tested positive both times.

- a) What is the probability that this patient has cancer? (**3 points**).
- b) What is the probability that this patient will test positive if they take the test the third time? (**3 points**)

Describe clearly the probability model you are using to calculate the above probabilities.

2. We have an urn with N balls of which θ are red and the remaining $N - \theta$ are white. We are assuming that N is known but θ is unknown. We sample n balls from the urn without replacement and suppose we see x red balls in the sample. What can we infer about θ based on knowledge of x , n and N ?

This problem has applications in survey sampling. Suppose we want to know the number θ in a city (of total population N) that support a particular political party. We take a sample of n people out of which x support that party. What can we then say about θ ?

- a) Argue that the likelihood function is given by: (**3 points**)

$$\mathbb{P}\{X = x \mid N, \theta, n\} = \frac{\binom{\theta}{x} \binom{N-\theta}{n-x}}{\binom{N}{n}}.$$

This is the Hypergeometric distribution.

- b) Work with the prior

$$\mathbb{P}\{\theta = j \mid N\} = \frac{1}{N+1} I\{j \in \{0, 1, \dots, N\}\},$$

and show that the posterior is given by (**3 points**)

$$\mathbb{P}\{\theta = j \mid x, N, n\} = \frac{\binom{j}{x} \binom{N-j}{n-x}}{\binom{N+1}{n+1}} I\{j \in \{0, 1, \dots, N\}\}.$$

c) Show that (**3 points**)

$$\mathbb{E}(\theta \mid x, n, N) = \frac{(N+2)(x+1)}{n+2} - 1$$

and

$$\text{var}(\theta \mid x, n, N) = \frac{p(1-p)}{n+3}(N+2)(N-n), \quad p := \frac{x+1}{n+2}.$$

d) When N is large, argue that (**2 points**)

$$\mathbb{E}\left(\frac{\theta}{N} \mid x, n, N\right) \approx \frac{x+1}{n+2} = p$$

and

$$\text{var}\left(\frac{\theta}{N} \mid x, n, N\right) \approx \frac{p(1-p)}{n+3}.$$

This implies that, when N is large, a point estimate of θ/N can be taken to be p and the uncertainty of this point estimate can be taken to be the posterior standard deviation $\sqrt{p(1-p)/(n+3)}$.

e) Let R_{n+1} denote the event that the next draw (i.e., the $(n+1)$ -th draw) leads to a red ball. Show that the posterior probability of R_{n+1} given x, n, N is given by (**3 points**)

$$\mathbb{P}\{R_{n+1} \mid x, n, N\} = \frac{x+1}{n+2} = p.$$

This formula is known as the Laplace Rule of Succession.

3. In class, we discussed the problem of estimating θ from numerical measurements $x_1, \dots, x_n$. We used the following Bayesian model:

$$\begin{aligned} X_1, \dots, X_n \mid \theta, \sigma &\stackrel{\text{i.i.d.}}{\sim} N(\theta, \sigma^2), \\ \theta, \log \sigma &\stackrel{\text{i.i.d.}}{\sim} \text{uniform}(-\infty, \infty). \end{aligned}$$

We proved that the posterior density of θ under this model satisfies

$$\frac{\sqrt{n}(\theta - \bar{x})}{\hat{\sigma}} \mid \text{data} \sim t_{n-1}, \tag{1}$$

where

$$\bar{x} := \frac{x_1 + \dots + x_n}{n}, \quad \hat{\sigma}^2 := \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

In this exercise, we shall go over a slightly different proof of the same result. Along the way, we shall also discuss inference for σ .

a) Show that

$$\theta \mid \text{data}, \sigma \sim N\left(\bar{x}, \frac{\sigma^2}{n}\right).$$

The left hand side above refers to the conditional distribution of θ given the data as well as the parameter σ . (**2 points**).

b) Show that the posterior density of σ satisfies (**2 points**)

$$f_{\sigma|\text{data}}(\sigma) \propto \sigma^{-n} \exp\left(-\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{2\sigma^2}\right) I\{\sigma > 0\}.$$

c) Use the results of the previous two parts along with

$$f_{\theta|\text{data}}(\theta) = \int_0^\infty f_{\theta|\text{data},\sigma}(\theta) f_{\sigma|\text{data}}(\sigma) d\sigma$$

to deduce (1). (**2 points**)

d) Use the result in part (b) to show that

$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2 \mid \text{data} \sim \chi_{n-1}^2. \quad (2)$$

On this basis, comment on whether

$$\hat{\sigma} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

is a reasonable point estimate of σ . (**2+1=3 points**)

e) Using (2), derive a $100(1 - \alpha)\%$ Bayesian credible interval for σ for fixed α . (**2 points**)

f) Suppose $n = 6$ and

$$x_1 = 26.6, \quad x_2 = 38.5, \quad x_3 = 34.4, \quad x_4 = 34, \quad x_5 = 31, \quad x_6 = 23.6.$$

Explicitly calculate the 95% credible interval for σ . (**2 points**)

4. Consider the frequentist probability

$$\mathbb{P}\left\{\bar{X}_N - \frac{t_{N-1,\alpha/2}}{\sqrt{N}} \sqrt{\frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X}_N)^2} \leq \theta \leq \bar{X}_N + \frac{t_{N-1,\alpha/2}}{\sqrt{N}} \sqrt{\frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X}_N)^2}\right\}, \quad (3)$$

where $X_1, X_2, \dots$ are i.i.d. $N(\theta, \sigma^2)$ for fixed θ and σ , and N is the stopping rule

$$N := \inf\{n \geq 1 : X_n \leq 25\}. \quad (4)$$

We discussed in class that, unlike the case when N is a fixed number, the probability (3) need not equal $1 - \alpha$. The goal of this problem is to verify this phenomenon by simulation.

Because (3) is not well-defined when $N = 1$, we instead consider the simpler probability

$$\mathbb{P}\left\{\bar{X}_N - \sigma \frac{z_{\alpha/2}}{\sqrt{N}} \leq \theta \leq \bar{X}_N + \sigma \frac{z_{\alpha/2}}{\sqrt{N}}\right\}. \quad (5)$$

Follow the steps below to estimate this probability.

a) Fix $\theta = 35.5$ and $\sigma = 5.5$.

b) Set $\alpha = 0.05$ and $M = 100000$.

c) For $i = 1, \dots, M$:

- Generate $N(\theta, \sigma^2)$ random variables $X_1, X_2, \dots$ until an observation less than 25 is obtained. Let N be the number of observations.
- Calculate

$$C_i = I \left\{ \bar{X}_N - \sigma \frac{z_{\alpha/2}}{\sqrt{N}} \leq \theta \leq \bar{X}_N + \sigma \frac{z_{\alpha/2}}{\sqrt{N}} \right\}.$$

d) Estimate (5) by

$$\frac{C_1 + \dots + C_M}{M}.$$

Write code and compute this estimate. Comment on how different the estimate is from $1 - \alpha$. **(5 points)**.

5. Consider again the problem of estimating an unknown physical quantity θ from data

$$x_1 = 26.6, \quad x_2 = 38.5, \quad x_3 = 34.4, \quad x_4 = 34, \quad x_5 = 31, \quad x_6 = 23.6.$$

Suppose that the measurement device can only record observations if they are in the range $[23, 39]$. Based on this information and the data, what can be inferred about θ ?

a) Argue that the likelihood now should be

$$\prod_{i=1}^n \frac{\frac{1}{\sigma} \phi\left(\frac{x_i - \theta}{\sigma}\right)}{\Phi\left(\frac{39 - \theta}{\sigma}\right) - \Phi\left(\frac{23 - \theta}{\sigma}\right)},$$

where

$$\phi(z) := (2\pi)^{-1/2} e^{-z^2/2}, \quad \Phi(z) = \int_{-\infty}^z \phi(u) du.$$

(2 points)

b) Using the same prior

$$\theta, \log \sigma \stackrel{\text{i.i.d.}}{\sim} \text{uniform}(-\infty, \infty),$$

write down the joint posterior density for θ and σ . **(2 points)**

c) Construct a dense grid of points $(\theta, \log \sigma)$, say in the range

$$[15, 50] \times [-1, 3].$$

On this grid, compute the unnormalized posterior density and normalize numerically. **(3 points)**

d) Based on the discretized posterior, compute an approximate 95% Bayesian credible interval for θ . Compare your interval to the interval obtained by the usual normal analysis from class which does not take the truncation into consideration. Does taking the truncation into account lead to a smaller or larger credible region? **(4 points)**

6. Consider again the problem of estimating an unknown physical quantity θ from data. Suppose the measurement device works correctly if the measurement lies in the range

[23, 39]. If the measurement falls outside this range, the device displays **Failure**. Suppose we observe

$$x_1 = 26.6, \quad x_2 = 38.5, \quad x_3 = 34.4, \quad x_4 = 34, \quad x_5 = 31, \quad x_6 = 23.6, \quad x_7 = \text{Failure}, \quad x_8 = \text{Failure}$$

Based on this information and the data, what can be inferred about θ ?

- a) Argue that the likelihood now should be

$$\left[\prod_{i=1}^6 \frac{1}{\sigma} \phi \left(\frac{x_i - \theta}{\sigma} \right) \right] \left[1 - \Phi \left(\frac{39 - \theta}{\sigma} \right) + \Phi \left(\frac{23 - \theta}{\sigma} \right) \right]^2.$$

(2 points)

- b) Using the same prior

$$\theta, \log \sigma \stackrel{\text{i.i.d.}}{\sim} \text{uniform}(-\infty, \infty),$$

write down the joint posterior density for θ and σ . **(2 points)**

- c) Construct a dense grid of points $(\theta, \log \sigma)$, say in the range

$$[15, 50] \times [-1, 3].$$

On this grid, compute the unnormalized posterior density and normalize numerically. **(3 points)**

- d) Based on the discretized posterior, compute an approximate 95% Bayesian credible interval for θ . Compare your interval to the interval obtained by the usual normal analysis from class which does not take the failures into consideration. **(4 points)**

7. Consider yet again the problem of estimating an unknown physical quantity θ from observations $x_1, \dots, x_n$. Now consider the following Bayesian model:

$$X_1, \dots, X_n \mid \theta, \sigma \stackrel{\text{i.i.d.}}{\sim} \text{Lap}(\theta, \sigma),$$

$$\theta, \log \sigma \stackrel{\text{i.i.d.}}{\sim} \text{uniform}(-\infty, \infty).$$

Here $\text{Lap}(\theta, \sigma)$ denotes the Laplace distribution with density

$$x \mapsto \frac{1}{2\sigma} \exp \left(-\frac{|x - \theta|}{\sigma} \right).$$

- a) Show that the posterior of θ satisfies **(4 points)**

$$f_{\theta|\text{data}}(\theta) \propto \left(\frac{1}{M(\theta)} \right)^n, \quad M(\theta) := \sum_{i=1}^n |x_i - \theta|.$$

- b) For the data

$$x_1 = 26.6, \quad x_2 = 38.5, \quad x_3 = 34.4, \quad x_4 = 34, \quad x_5 = 31, \quad x_6 = 23.6,$$

evaluate the density at a dense grid of points in some large range, for example $[15, 50]$, normalize correctly so that the integral of the posterior density is 1, and plot the posterior density. Comment on the nature of the plot. **(4 points)**

- c) Use your discretized approximation of the posterior to construct an approximate 95% credible interval for θ . Compare your interval with the interval derived from the normal-based method from class. (**4 points**)
- d) Repeat part (c) when the data are

$$x_1 = 26.6, \quad x_2 = 38.5, \quad x_3 = 34.4, \quad x_4 = 34, \quad x_5 = 31, \quad x_6 = 23.6, \quad x_7 = 120.$$

The additional observation $x_7 = 120$ is an outlier. Compare how this outlier affects the intervals obtained from the normal and Laplace models. (**3 points**)